

DFSMS Basics: Introduction to DFSMSHsm

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March 6th, 2006, S3015
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Agenda



- Introduction to DFSMShsm
- DFSMShsm Architecture
- Data Set Basics
 - ▶ Data Set Life Cycle
 - ▶ Basic Catalog Structure
- DFSMShsm Functions
 - ▶ Space Management
 - ▶ Availability Management
 - ▶ Other Support Functions

What is DFSMShsm?



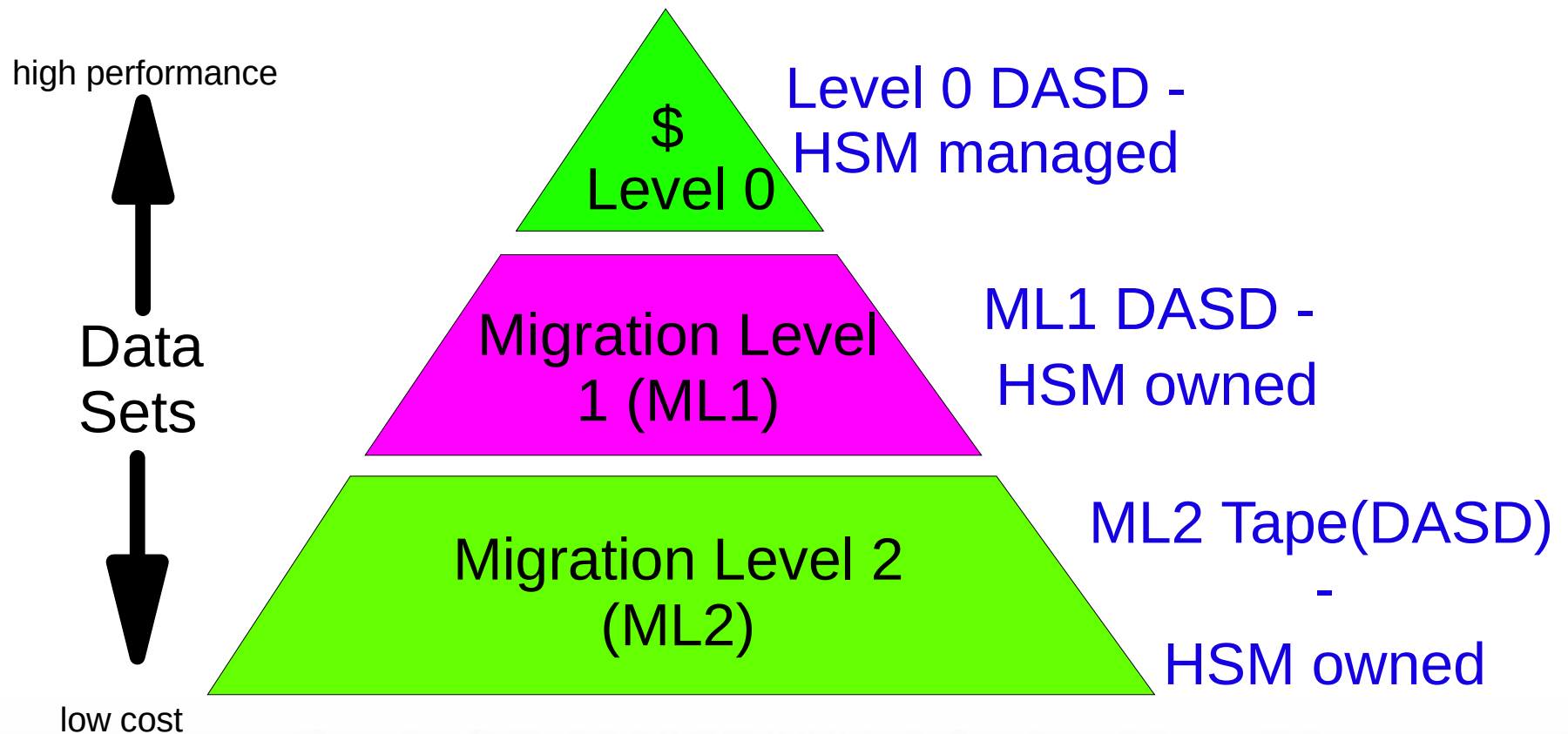
Hierarchical Storage Manager

- To manage storage resources based on a hierarchy of cost and service-level policies

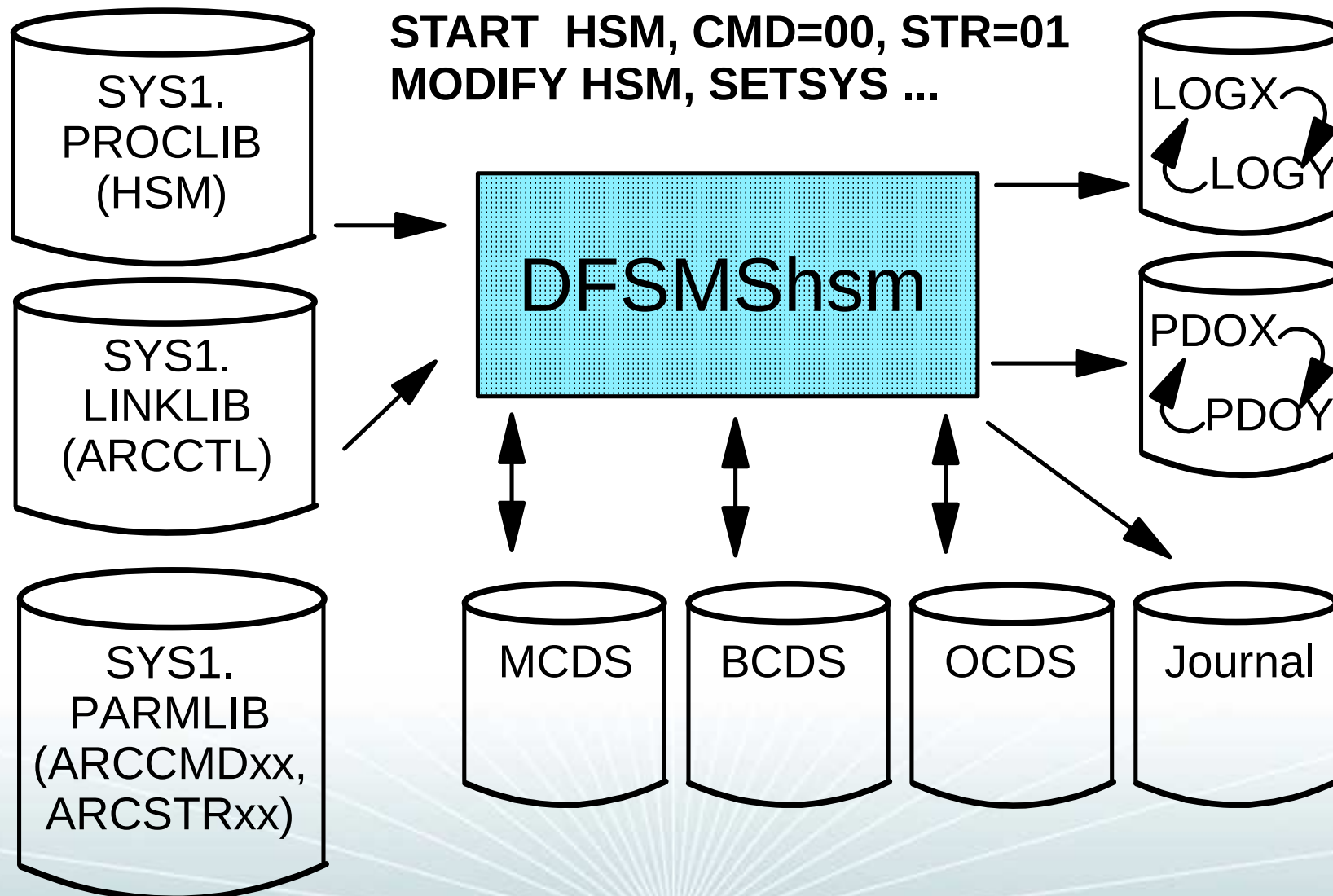
DFSMShsm is an element of z/OS[®] (formerly OS/390[®] and MVS[™])

- Major Functional Areas
 - ▶ Space Management
 - ▶ Availability Management
 - ▶ Various Support Functions

DFSMShsm Storage Hierarchy



Behind the Scenes



DFSMSHsm Architecture



- DFSMSHsm is a "started task"
 - ▶ This means that DFSMSHsm is started on a z/OS system and remains operating in the system until it is stopped, canceled or abnormally ends.
- Every "*started task*" is associated with a procedure in a procedure library.
 - ▶ DFSMSHsm is started by issuing an MVS **Start** command
 - ▶ Issue "**S HSM....**" on the system console, where "HSM" is the name of the procedure in SYSn.PROCLIB
 - ▶ The DFSMSHsm procedure contains information that DFSMSHsm requires at startup to properly configure itself

DFSMSHsm Architecture



- DFSMSHsm is executed as a "Load Module"
 - ▶ **ARCCTL** - DFSMSHsm load module name
 - ▶ Typically resides in SYS1.LINKLIB or equivalent
 - ▶ ARCCTL is link-edited with AC(1) so it can run authorized

- The DFSMSHsm proclib contains
 - ▶ Name of DFSMSHsm load library member
 - EXEC PGM=ARCCTL
 - ▶ DD statements for important data sets that DFSMSHsm uses
 - ▶ Various startup command options

DFSMSHsm Architecture



- DFSMSHsm works with startup member(s) located in SYSn.PARMLIB
 - ▶ Member name is **ARCCMDxx**, where
 - "xx" is a number between 00 and 99
 - e.g., **ARCCMD01**
 - ▶ ARCCMDxx member contains DFSMSHsm commands that you want issued during DFSMSHsm initialization processing every time you start DFSMSHsm. For example:
 - **SETSYS** commands to set DFSMSHsm system settings
 - **DEFINE** command to define various DFSMSHsm settings
- ▶ If you are not using MVS concatenated PARMLIB support, member ARCCMDxx must be in the data set specified with DD statement HSMPARM in the startup procedure.

DFSMSHsm Architecture



- ▶ Another parmlib member name is **ARCSTRxx**, where
 - "xx" is a number between 00 and 99
 - e.g., **ARCSTR01**
- ▶ If you need to use more startup procedure keywords than can be accommodated by the **100-character PARM limit** as detailed in z/OS MVS JCL Reference under the EXEC parameter
 - Use the **STR=xx** keyword within the PARM keywords to create a PARMLIB member ARCSTRxx to contain the remaining startup parameters.
 - As with member ARCCMDxx, if you are using MVS concatenated PARMLIB support, members ARCCMDxx and ARCSTRxx need not be in the same PARMLIB data set.

DFSMSHsm Architecture



- DFSMSHsm's metadata is maintained in 4 disk data sets
 - ▶ Migration Control Data Set (**MCDS**)
 - Contains information about migrated data sets and migration volumes
 - ▶ Backup Control Data Set (**BCDS**)
 - Contains information about backed up data sets and backup volumes
 - ▶ Offline Control Data Set (**OCDS**)
 - Contains information about tapes used by DFSMSHsm
 - ▶ **Journal**
 - Contains an entry to track every DFSMSHsm CDS Update
 - Used to forward recover a CDS
 - ▶ All CDSs are VSAM KSDS data sets

DFSMSHsm Architecture



- More DFSMSHsm CDS and Journal Information
 - ▶ CDSs can be VSAM RLS or non-RLS (Record Level Sharing)
 - RLS allows VSAM data sets to be shared between DFSMSHsm hosts at a record level rather than at a volume level
 - ▶ CDSs can be extended format or non-extended format
 - Extended format allows individual cluster to be larger than 4-gigabytes
 - ▶ MCDS and BCDS can be up to 4 clusters
 - ▶ OCDS can only be 1 cluster
 - ▶ Journal can be normal sequential data set or large format sequential data set
 - Must be allocated as single-volume and single extent

DFSMSHsm Architecture



■ Other data sets used by DFSMSHsm

▶ PDOx/PDOy datasets

- Optional, but *highly recommended*
- Contain trace entries that trace DFSMSHsm internal activity
- Tracing facility is called PDA - Problem Determination Aid

▶ LOGx/LOGy datasets

- Optional, unless you have certain OEM products.
- Contain meta-data logged by various DFSMSHsm functions
- Often used by OEM products to track DFSMSHsm activity



DFSMSHsm Architecture



■ 2 Levels of DFSMSHsm Commands

► **Storage Administrator** Commands

- These commands should be limited to a few Storage or Security Administrators
- Authorization to these commands can be controlled by RACF Facility Class profiles
- Described in **DFSMSHsm Storage Administration Reference** manual
- Can be issued from TSO via the HSEND CMD command

► **End-User** Commands

- Small set of DFSMSHsm commands that can be issued by end-users
- Authorization to these commands can be controlled by RACF Facility Class profiles
- Described in **DFSMSHsm Managing Your Own Data** manual
- Can be issued from TSO via the various H..... commands (e.g. HMIGRATE)

DFSMSHsm Architecture



- DFSMSHsm can receive commands from
 - ▶ Operator's Console
 - Via MVS Modify Command.
 - ***F HSM,SETSYS TAPEMIGRATION(DIRECT)***
 - ▶ TSO
 - Issued from TSO terminal by an end-user or Storage Administrator
 - ***HMIGRATE USER.DATASET***
 - ***HSEND MIGRATE DSN(USER.DATASET)***
 - ▶ ISMF Panels
 - Various panels allow issuing various DFSMSHsm commands
 - ▶ Batch Jobs
 - ***EXEC PGM=IKJEFT01***
 - commands listed in SYSTSIN stream

DFSMSHsm Architecture



- DFSMSHsm can also receive commands from
 - ▶ ARCCMDxx member at DFSMSHsm initialization
 - Any DFSMSHsm authorized command can be put into this PDS member
 - ▶ Programmatically via various DFSMSHsm assembler macros
 - e.g.. ARCHBACK or ARCHSEND
 - ▶ Also there are some DFSMSHsm programs that you can put in your JCL stream
 - ARCINBAK - allows initiation of DFSMSHsm backup commands from a batch job

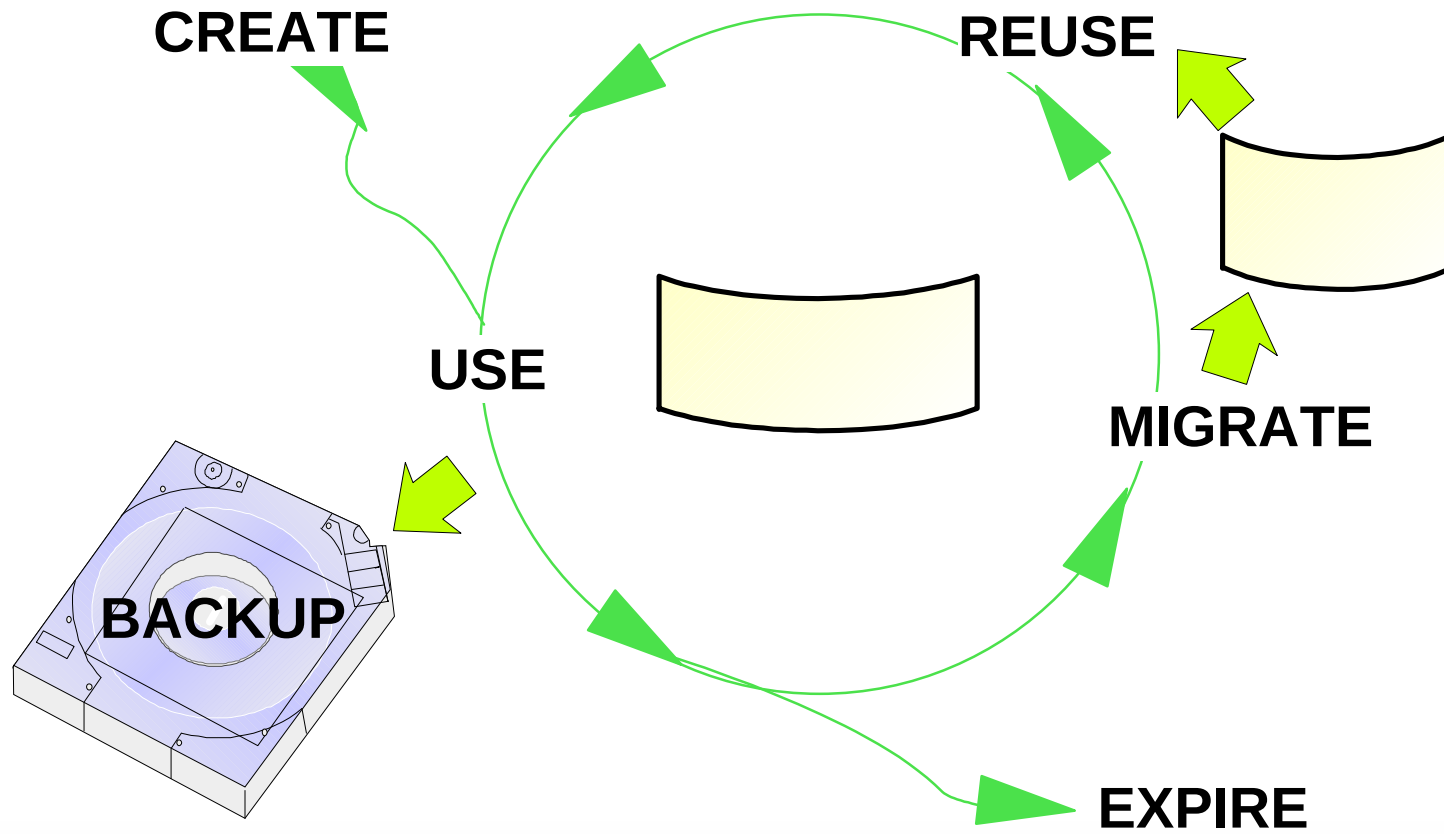


DFSMSHsm Architecture

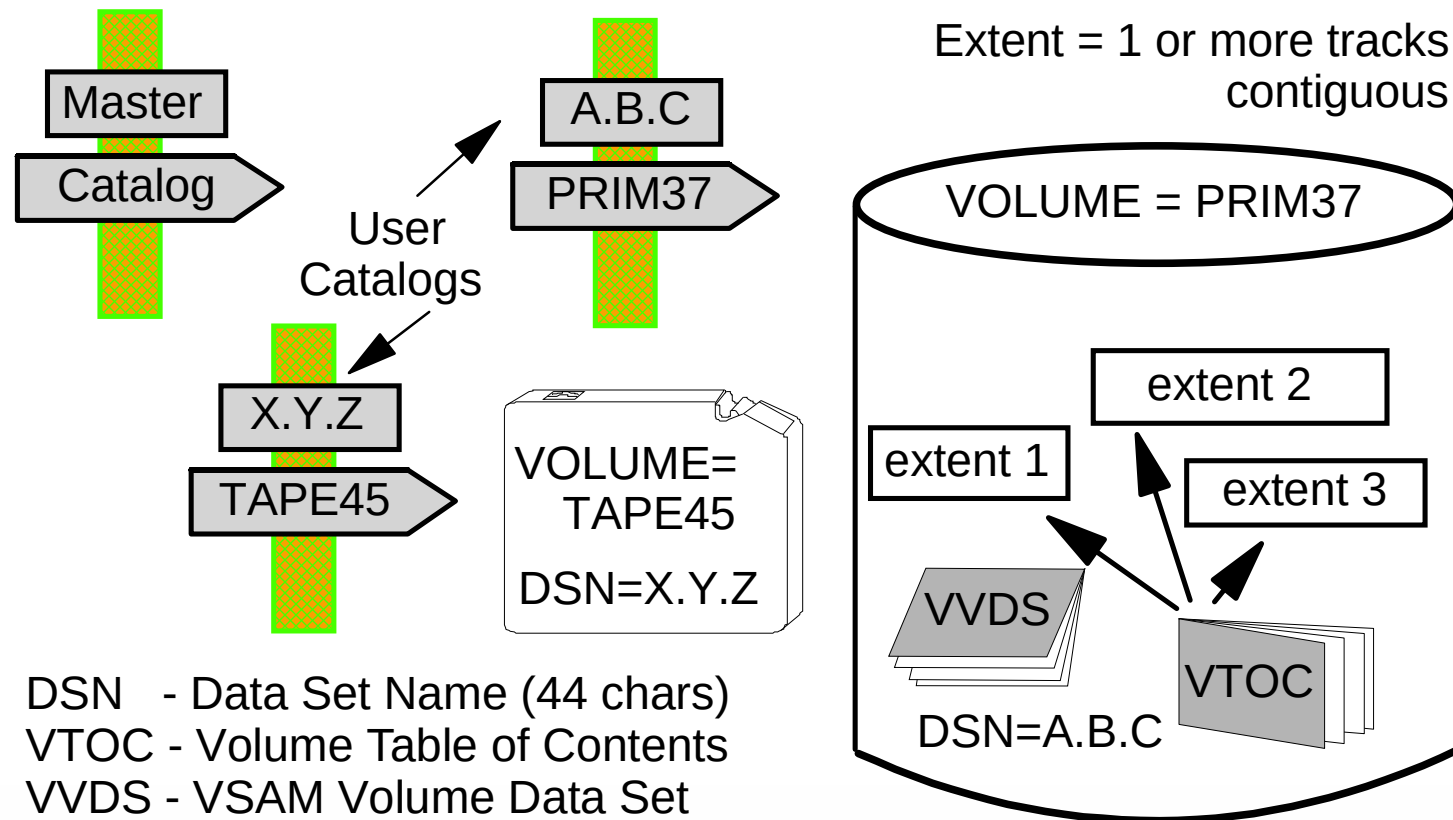


- DFSMSHsm basically allows 3 methods for initiating various processes
 - ▶ Scheduled Functions
 - Days grouped into cycles (which days to run the process)
 - Times determined by START/STOP windows
 - Process automatically starts on appropriate days/time of day
 - ▶ Event-Driven Functions
 - Functions initiated when a certain event occurs such as a timer-pop
 - ▶ Command-Initiated Functions
 - Storage Administrators, Security Administrators or End-Users issue a command

Life Cycle of a Dataset



Basic Catalog Structure



DFSMSHsm Space Management



- **Goal:** To ensure adequate free space exists on critical disk devices so that allocations can occur when needed
 - ▶ Aids in meeting service-level agreements for disk space
 - ▶ Prevents out of space conditions that could negatively affect mission critical applications

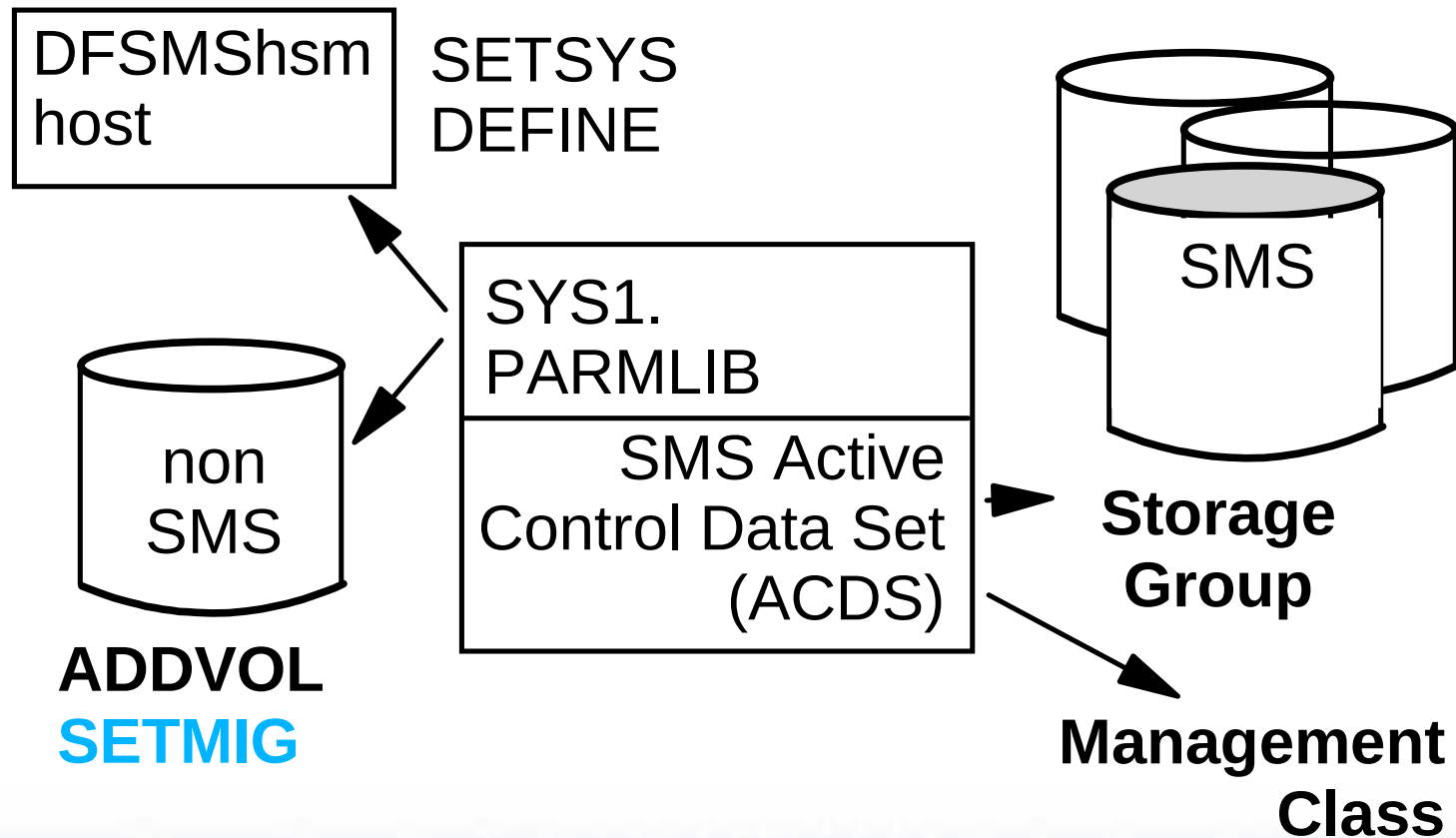


DFSMSHsm Space Management

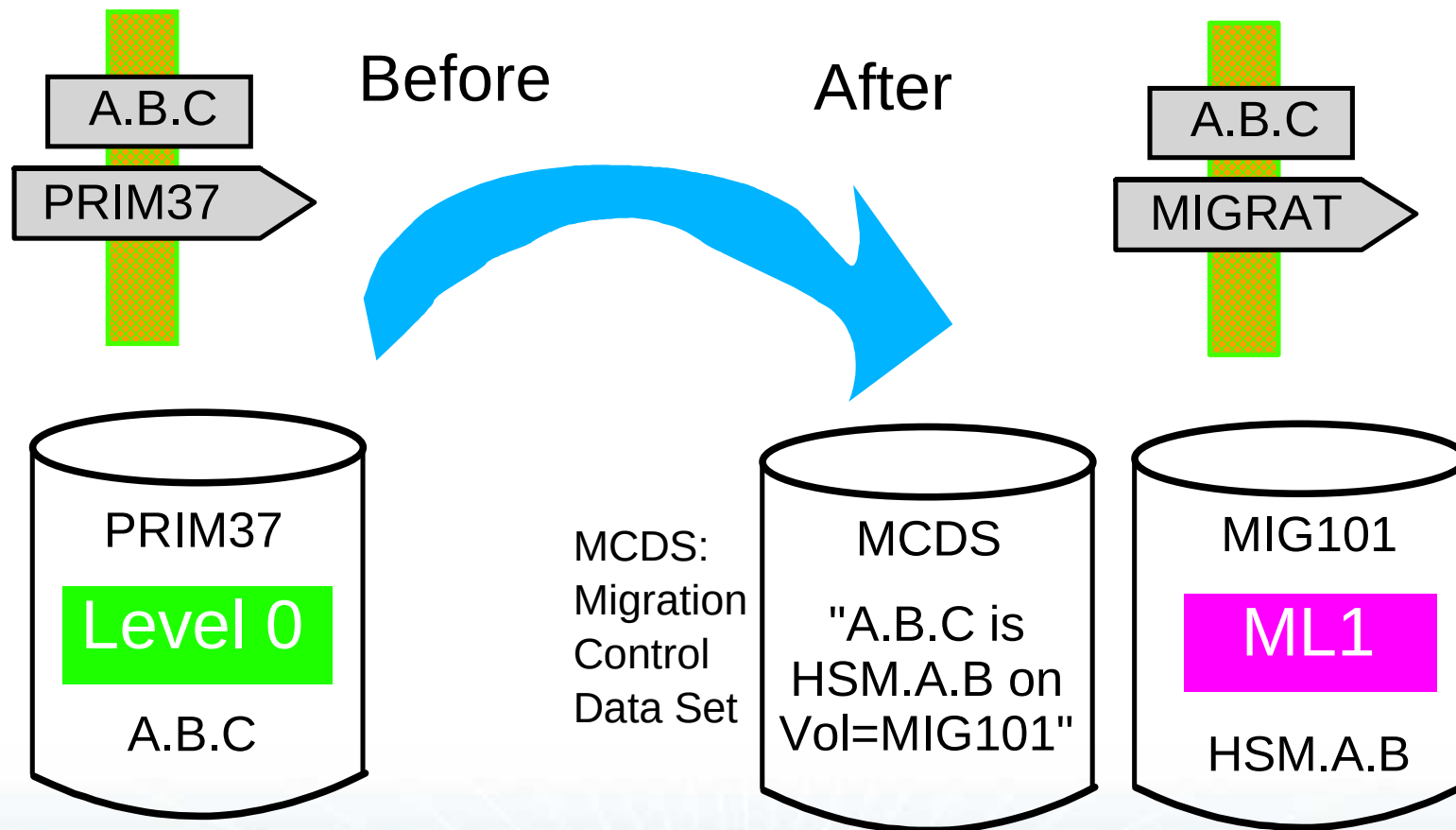


- Manages disk storage efficiently by:
 - ▶ Expiring data sets that have passed their expiration criteria (**Expiration**)
 - ▶ Moving unreferenced data to where it can be stored less expensively (**Migration**)
 - ▶ Compacting or compressing data as it is moved to new medium
 - **Compaction - software**
 - **Compression - hardware**
 - ▶ Freeing unused portions of data set (**Partial Release**)
 - ▶ Returning migrated data to user disk when needed (**Recall**)
 - PDSs are compressed when recalled

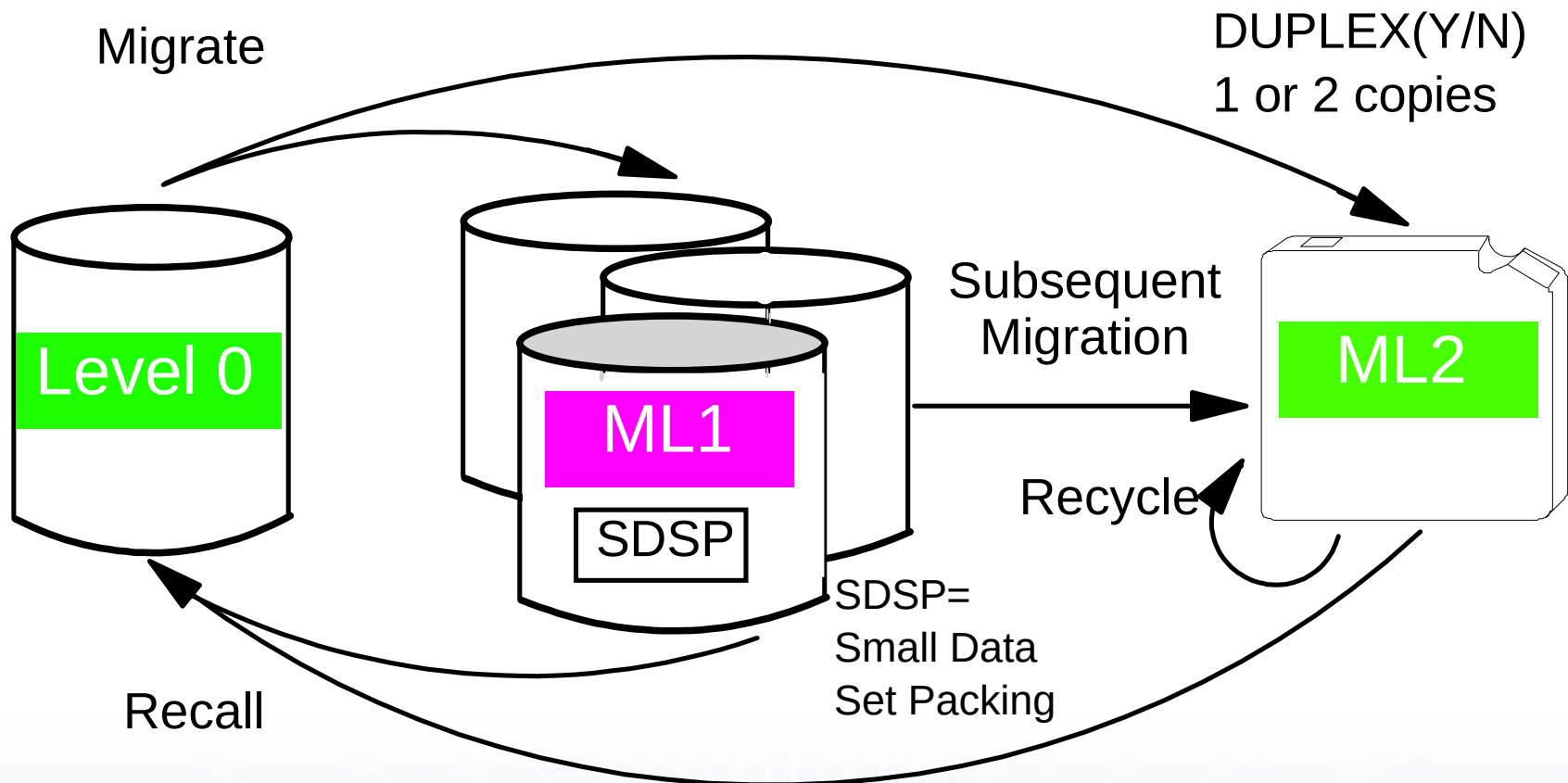
DFSMSHsm Policies for Space Management



Anatomy of a Migrated Data Set



Migrated Data Flow



DFSMSHsm Space Management



■ Migration Functions:

- ▶ ***Automatic Primary Space Management*** - Normally is processed once a day and attempts to process all eligible Primary Volumes
- ▶ Each primary volume is processed in two passes
 - **Pass 1**
 - Temporary data sets are deleted
 - Expired data is deleted
 - Overallocated space is released
 - If Fast Subsequent Migration is turned on, FSM eligible data sets are reconnected
 - Determines which data sets on the volume are candidates for migration or extent reduction.
 - **Pass 2** - if volume still over its low threshold of occupancy
 - Processes candidates for extent reduction
 - Migrates eligible data sets

DFSMSHsm Space Management



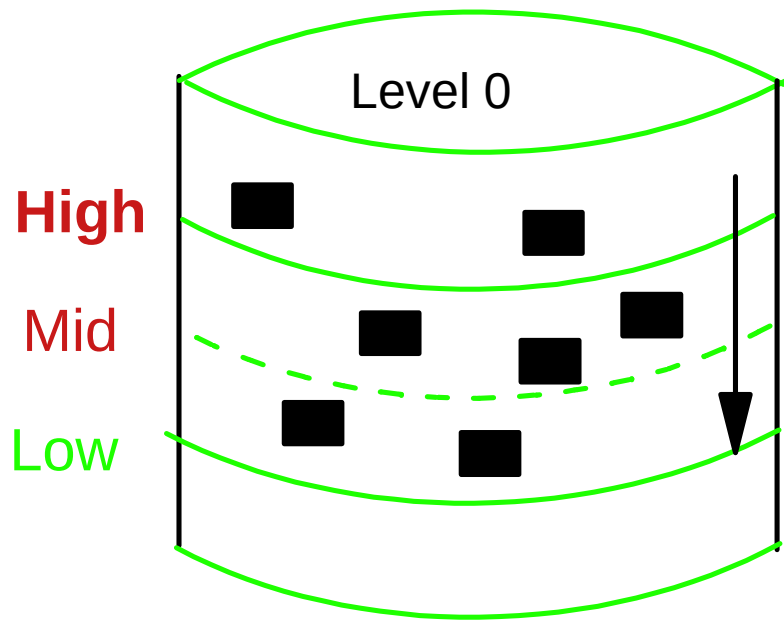
■ Migration Functions:

- ▶ **Automatic Secondary Space Management** - Normally is processed once a day and has several functions
 - Delete expired **migrated** data sets
 - Cleans up expired migration records from MCDS
 - Performs migration from **ML1 to ML2** for eligible data sets

- ▶ **Interval Migration**
 - Normally runs hourly and processes all eligible Primary Volumes

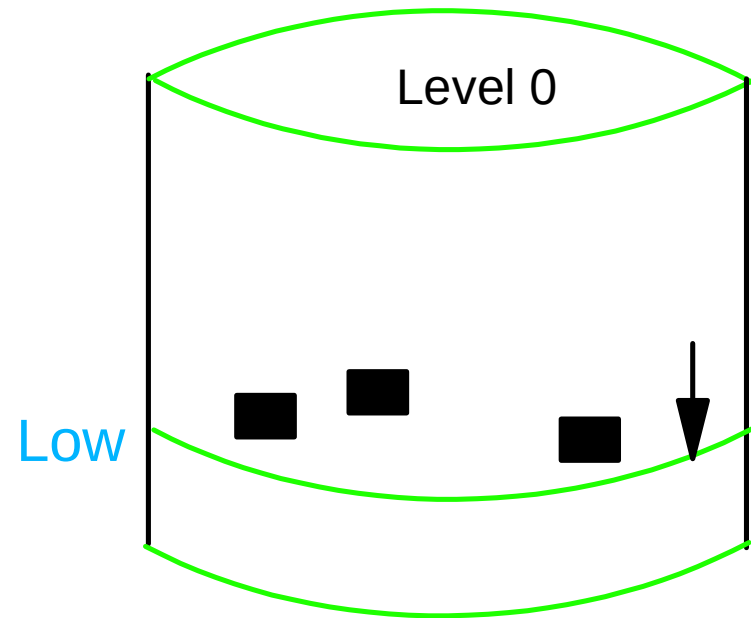


Space Management Thresholds



Interval Space Management
(Process Hourly)

Trigger is either High or Midpoint



Primary Space Management
(Process Once per Day)

Any volume above low threshold is
processed for migration

Recall Processing

Recall of data sets back to user volumes

- Event-driven (on reference to data set)
- Can also be performed by command

Recall volume selection

- SMS - ACS routines directs data set to storage group
- NON-SMS - Target volume can be specified on command, otherwise selected by DFSMSHsm ADDVOL parameters



Availability Management



Why?

- Help protect corporate information assets from loss

How?

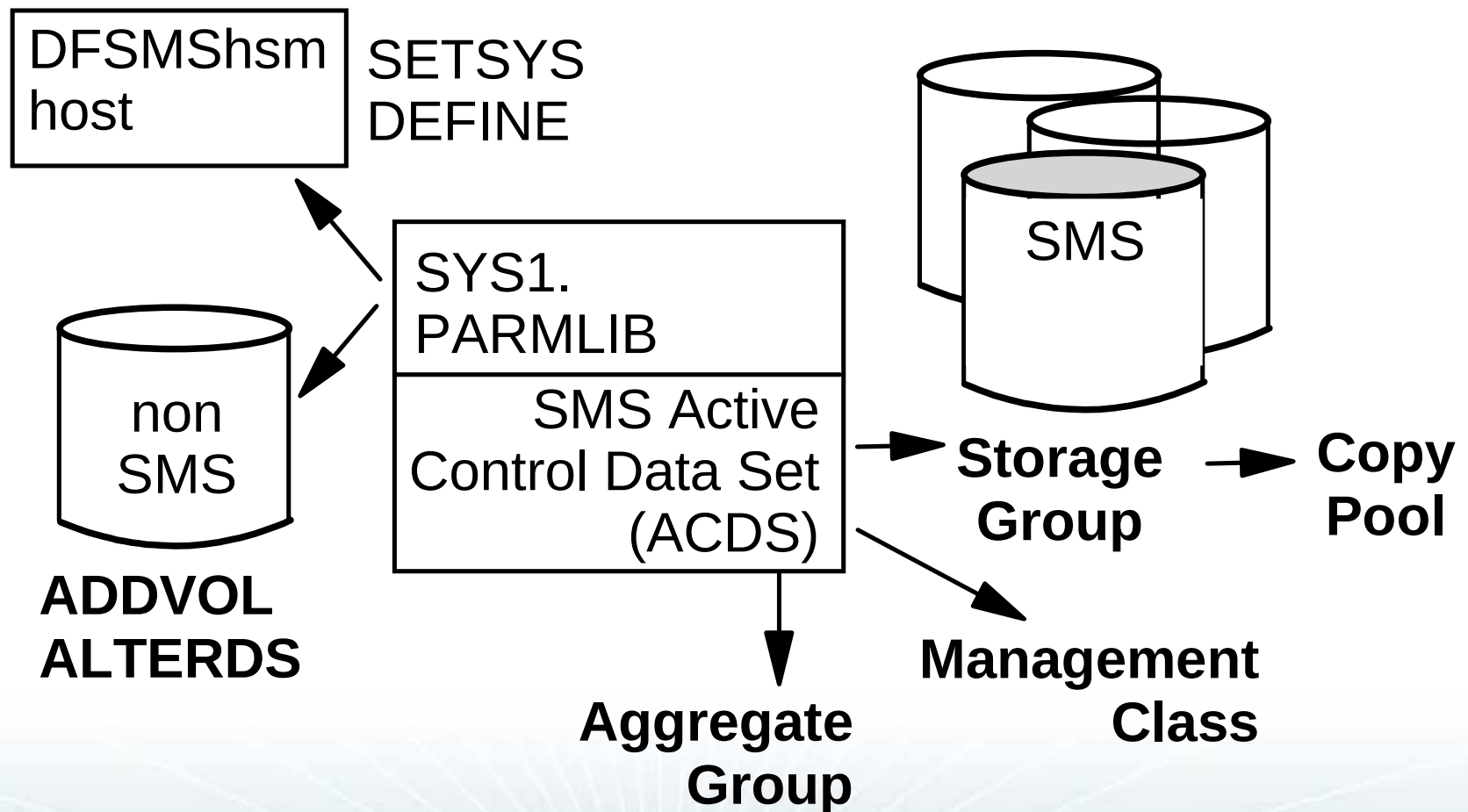
- Backup of Control Data Sets
- Backup of individual data sets
- Backup of DASD volumes
- Backup of aggregate groups of data
- Fast Replication backup of Copy Pools



Backup Policies

- Backup even if not changed or only backup if changed
- Frequency of Backup
 - ▶ Time between backups even if data changed.
- Guaranteed Backup Frequency
 - ▶ No more than this time between backups even if NOT changed.
- How Many Backup versions to keep
 - ▶ Up to 100 versions
- How long to keep backup versions

Policies for Availability Management



Scheduled Functions



Automatic incremental backup

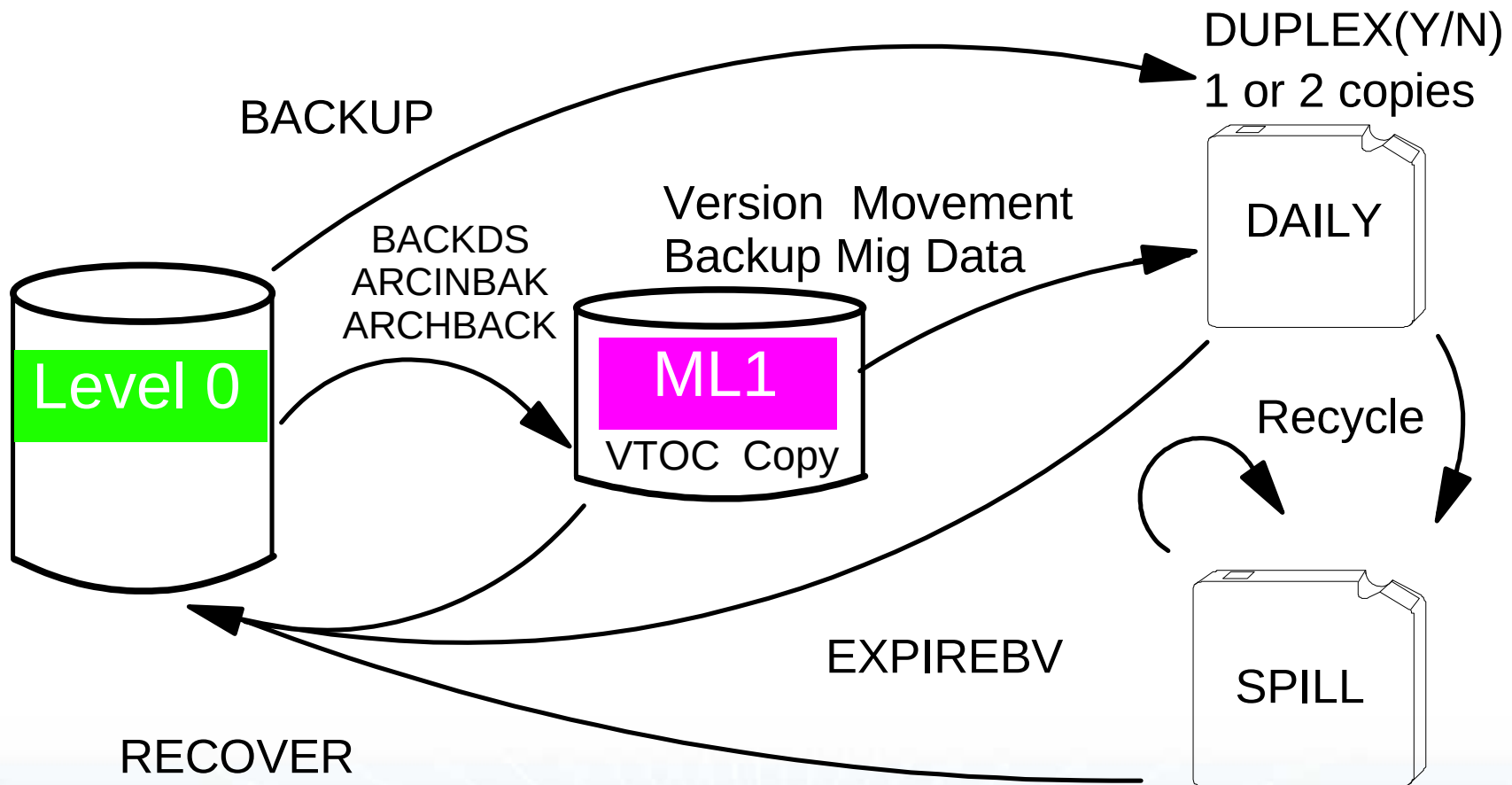
- Backup Control data sets and journal data
- Move backup versions from ML1
- Backup migrated data sets
- Backup user volumes

Automatic dump

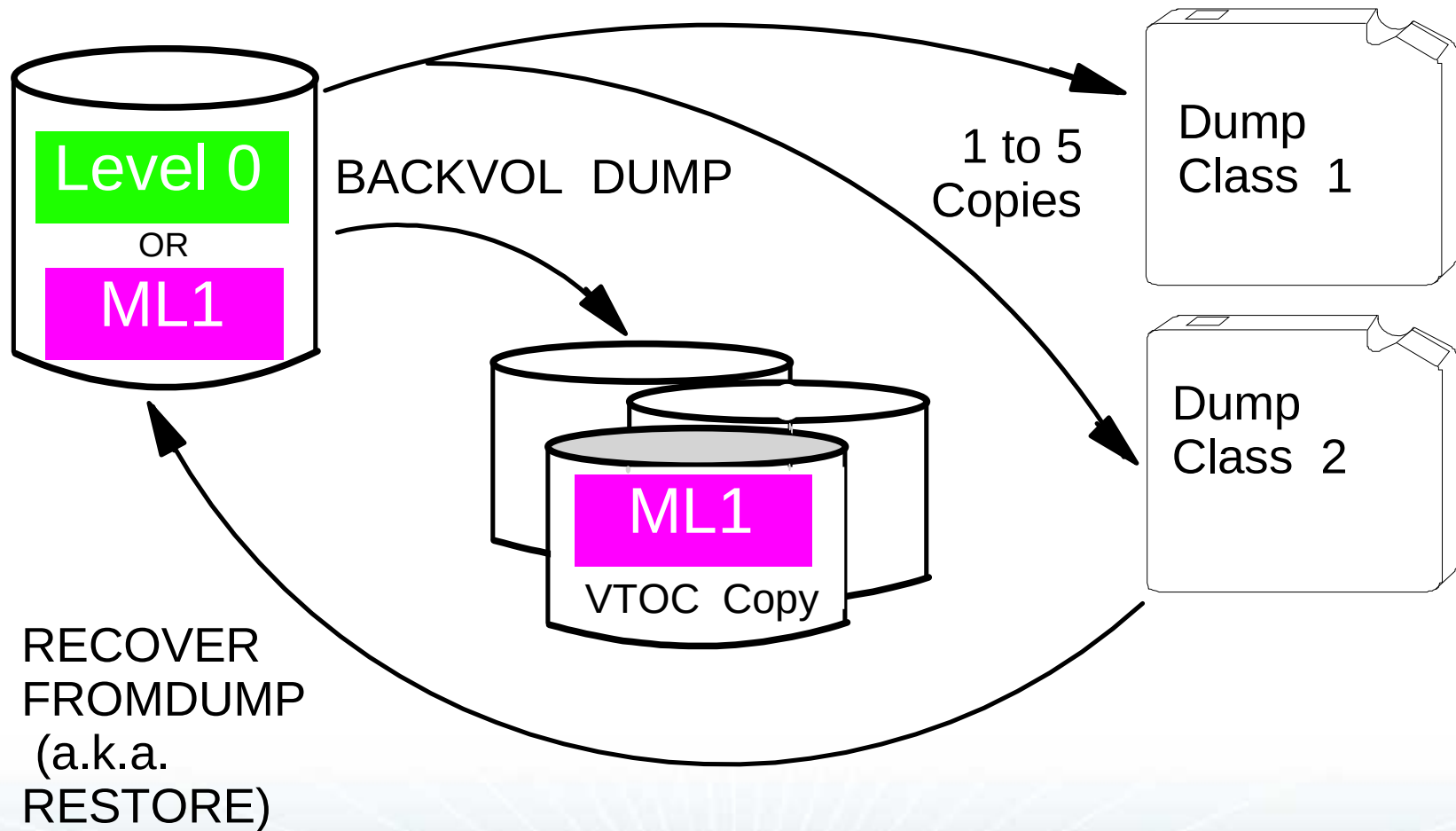
- Expire dump tapes
- Dump user volumes
- Delete excess VTOC copy data sets



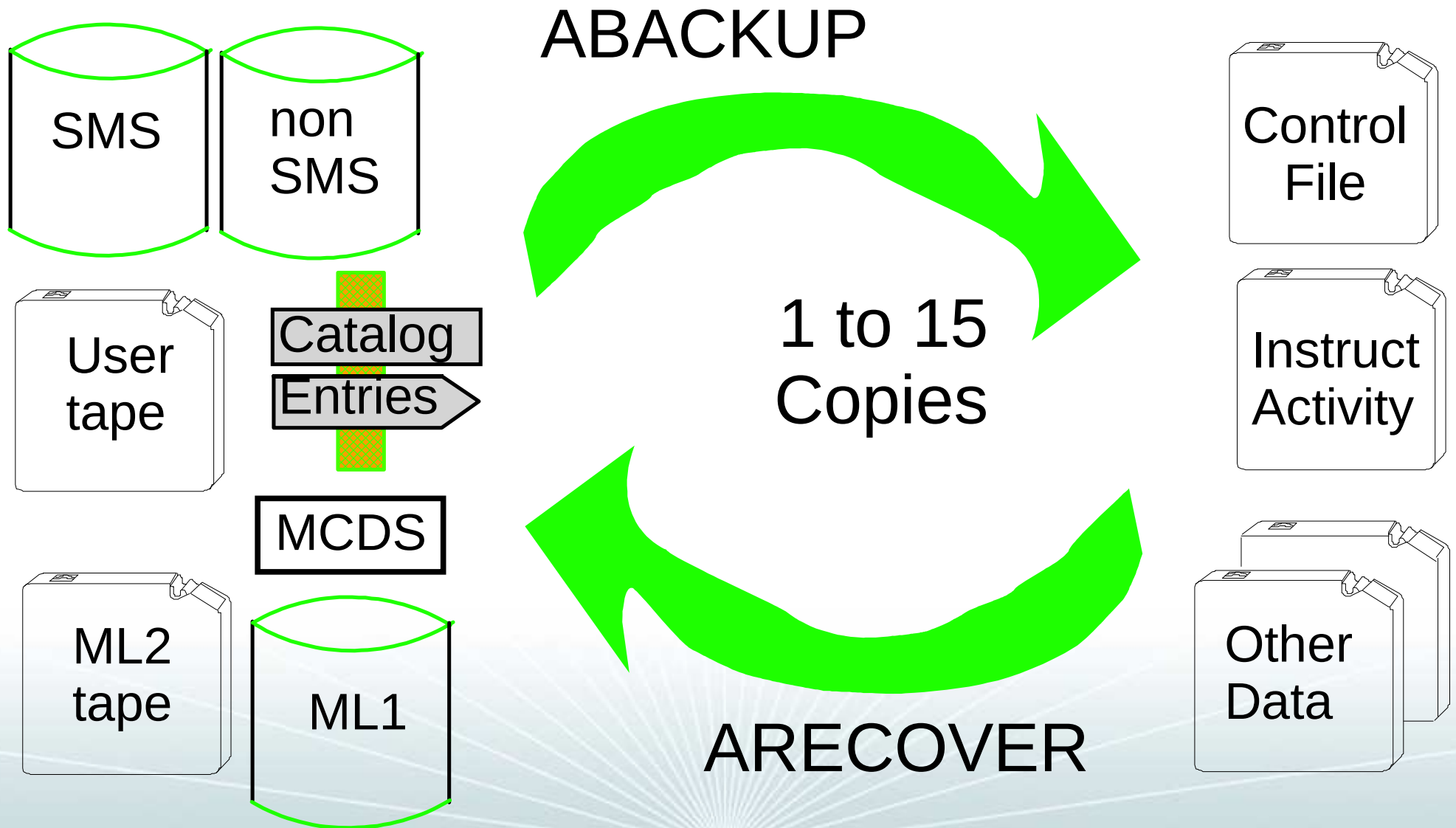
Backup Data Flow



Dump Data Flow



Aggregate Backup and Recovery Support (ABARS)

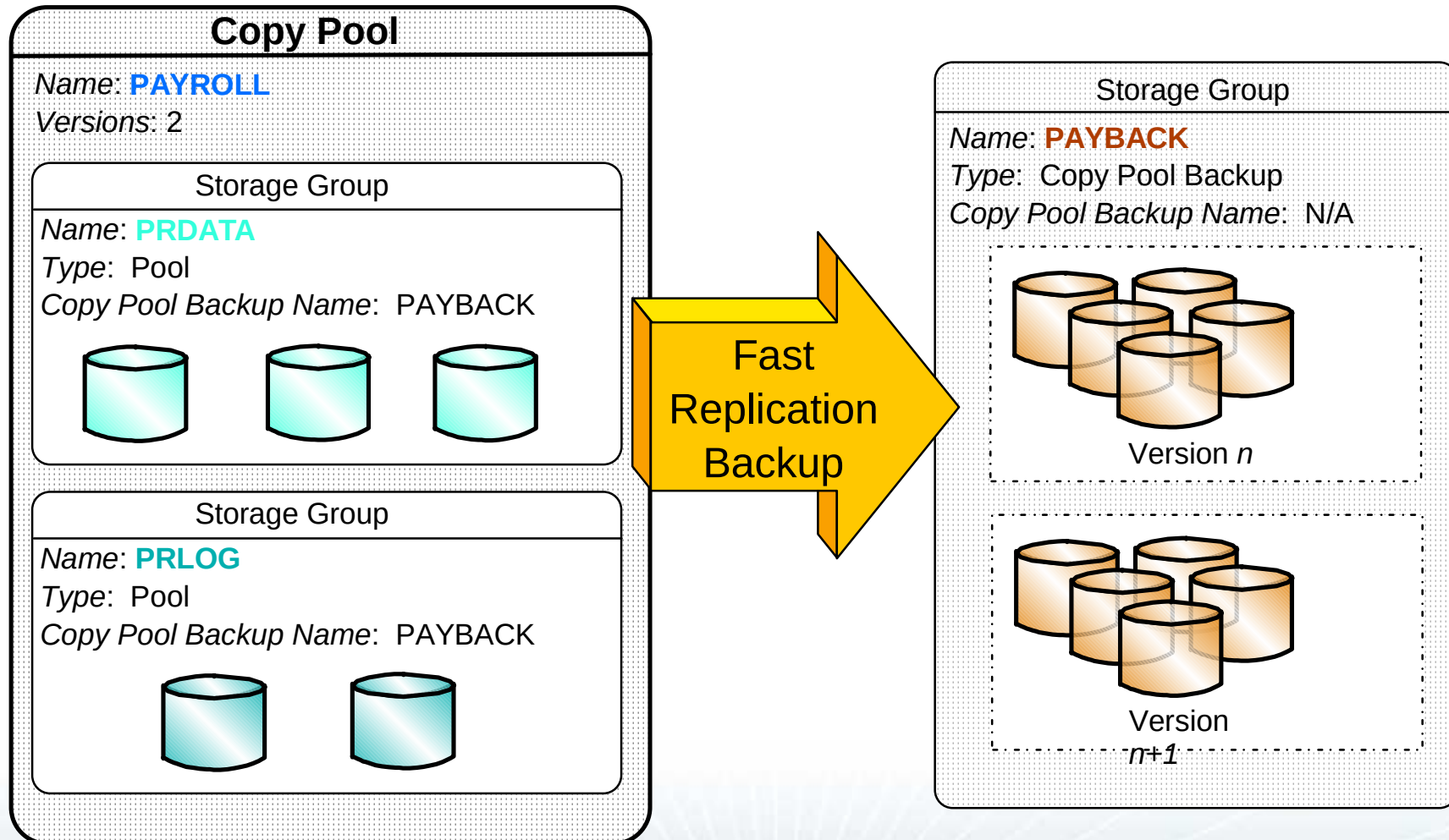


Fast Replication Overview



- DFSMSHsm invokes volume-level FlashCopy to create backup versions for sets of volumes
 - ▶ Serialized backup of hundreds of volumes in a matter of minutes
 - ▶ DFSMSHsm V1R5
 - ▶ Works with DB2 V8 System Point-in-Time Recovery
- Implementation is not limited to FlashCopy
 - ▶ Snapshot function is also supported
 - ▶ Will support any future fast replication functions supported by DFSMSdss
- Generic terminology of **"Fast Replication"** is used
 - ▶ FlashCopy refers to Fast Replication function supported by IBM ESS disk and HDS disk
 - ▶ Snapshot refers to the Fast Replication function supported by IBM RVA and STK SVA.

Fast Replication Overview



Administration



Why?

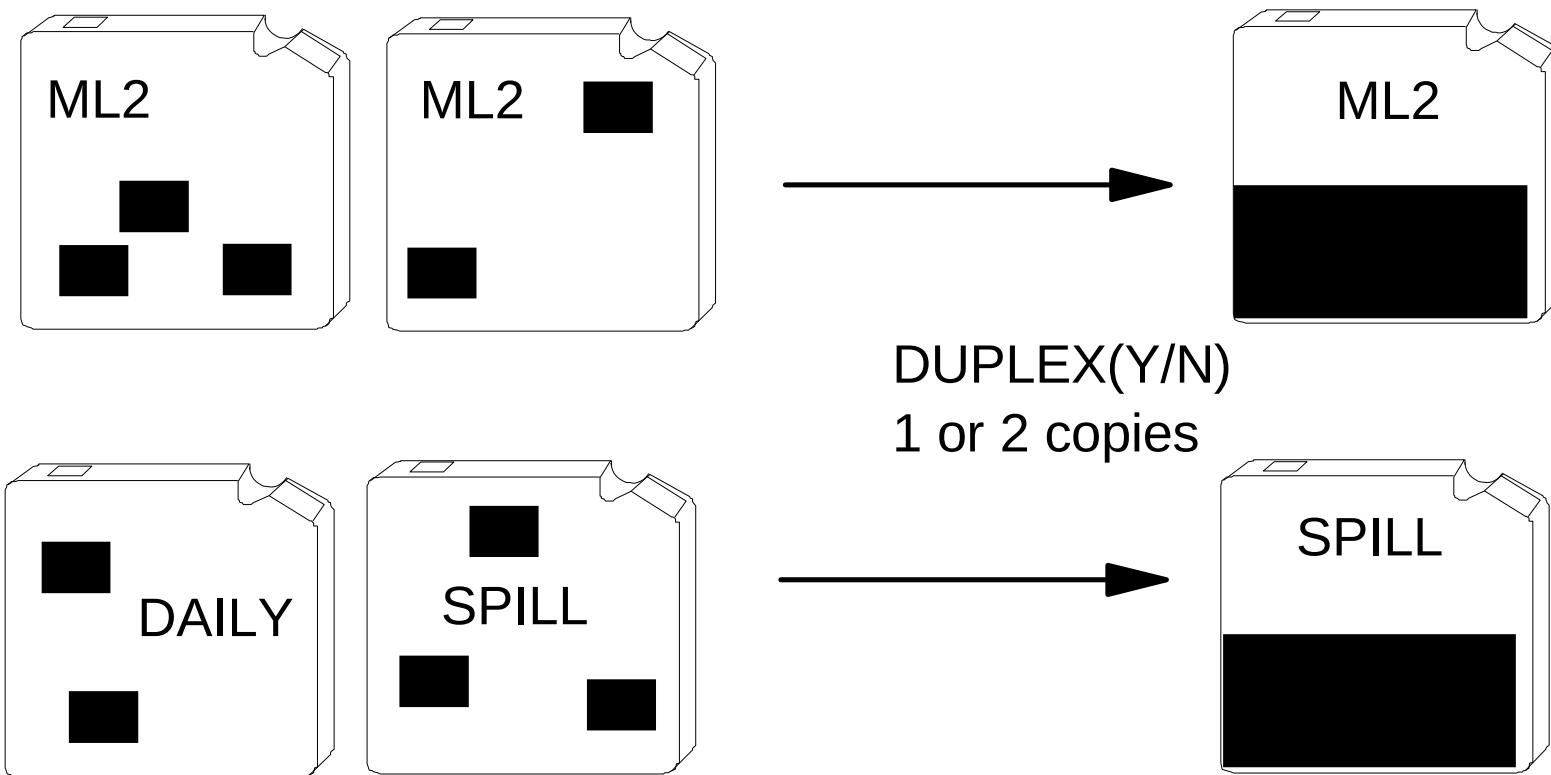
- Assist with the administration of DFSMSHsm

How?

- Movement of DFSMSHsm data on DASD
- Movement of DFSMSHsm data on tape
- Making duplicate tape copies
- Reporting functions

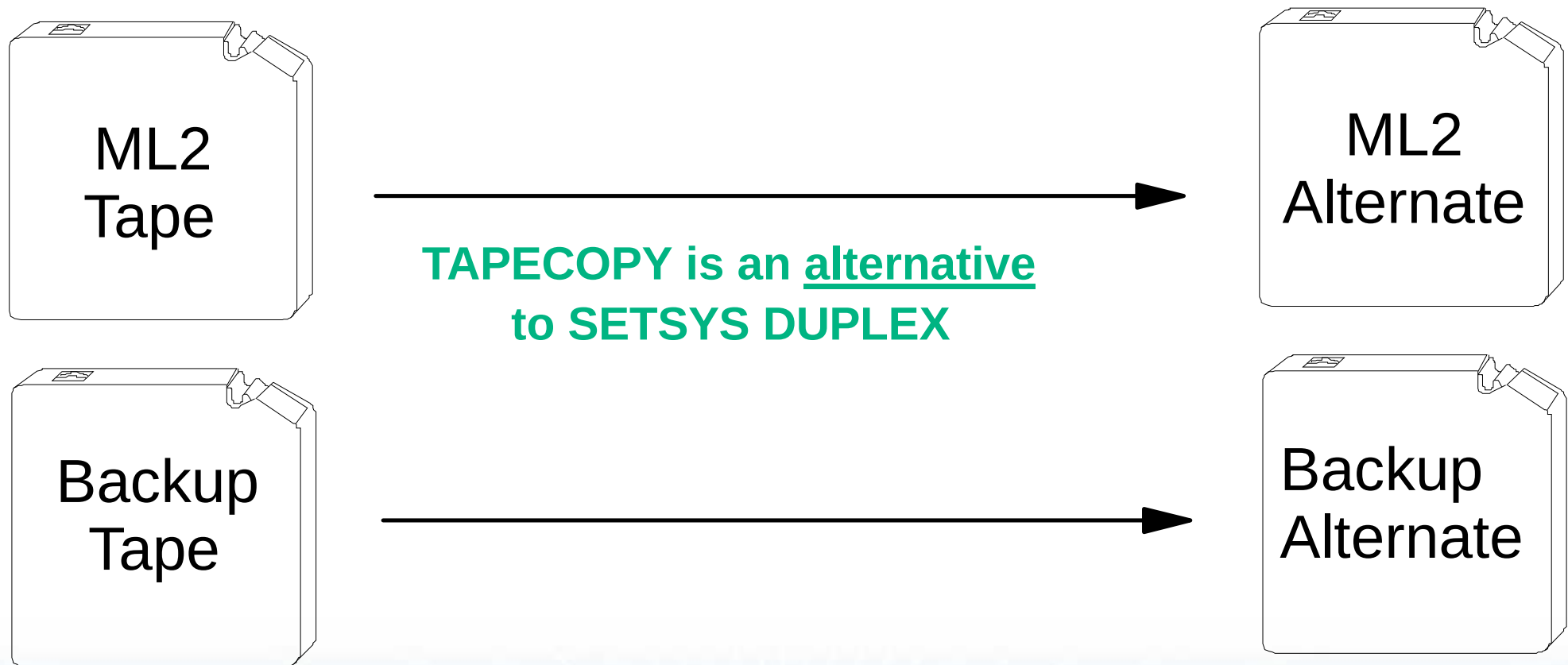


Recycle - Consolidate tape data



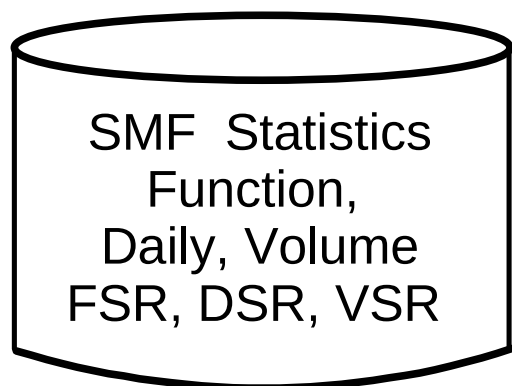
Input tapes are returned to scratch for reuse

TAPECOPY - Duplicating Tapes



TAPEREPL - Use alternate instead of Original Tape

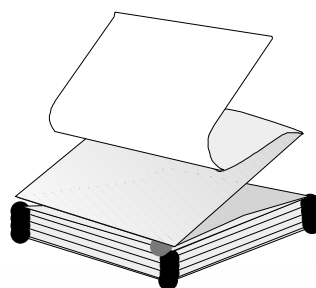
Reporting by DFSMShsm



QUERY

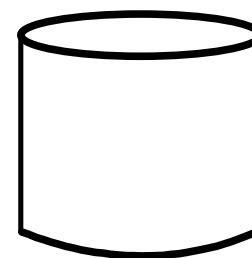
Activity Logs

- Backup
- Migration
- Dump
- Command
- ABACKUP
- ARECOVER



SYSOUT

AUDIT
LIST
REPORT



OUTDATASET

References - Publications



DFSMS/MVS General Information (GC26-4900)

DFSMSHsm Implementation and Customization Guide (SH21-1078)

DFSMSHsm Storage Administration Guide (SH21-1076)

DFSMSHsm Storage Administration Reference (SH21-1075)

DFSMSHsm Managing Your Own Data (SH21-1077)

DFSMSHsm Data Recovery Scenarios (GC35-0386)

DFSMSHsm Primer (SG24-5272) - ITSO Redbook

DFSMSdfp Storage Administration Reference (SH26-4920)

Summary



DFSMSHsm provides

- Automated space management
- Automated availability management
- Command and Application interfaces
- Disaster Recovery protection
- Reporting Capability

